

PEDRO RAMOS

Software Engineer

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MAJOR EXPERIENCES

Compilers Professor - PUC Minas University

Sep 2023 – Today

- 2 Years teaching compilers, operating systems and parallel computing in a renowned university in Brazil.
- Familiarity with bytecode (js, python and LLVM's), virtualization, optimization, lattice theory, flow analysis.
- OS principles, system calls, POSIX commands, distributed systems.
- Foundation on CUDA and OpenMP.
- Relevant project: Smart Contracts security analysis - finding tainted flow statically and dynamically in smart contracts written in Solidity.
- Relevant project: Operator Fusion in every ML Model: finding new fusions with reinforcement learning through thorough comparison between all ML models.

Sr. Consultant - Microsoft

Sep 2023 – Apr 2024

- Multicloud solutions using AKS + Cosmos DB. Java 17

Software Engineer - Paradigm

May 2022 – Feb 2023

- Crypto ecosystem (DeFi)
- Go & Python. Requests to Starknet/Cairo (L2) and Ethereum/Solidity (L1).

Software Engineer - B2W

Mar 2021 – Jul 2021

- Largest e-commerce of Latin America
- DDD web backend in go. concurrency everywhere.

Software Engineer - TremTec

Apr 2020 – Oct 2020

- Software factory start-up
- Python3 web-crawler ETL

Software Engineer - Take Blip

Sep 2019 – Apr 2020

- Chatbots in Node and .netcore . Hosted in Azure

Software Engineer (Mid-level) - SmarttBot

Aug 2018 – Aug 2019

- Python & Go. | Daytrading - highly efficient and responsive software.

Fullstack Engineer (Entry-level) - dti digital technologies

Feb 2018 – Jul 2018

- Software factory. First real job in the industry. Junior Dev. Angular and .net-core.

Compilers Engineer (Internship) - LG Electronics

2016 – 2018

- Implemented TaskMiner, a source-to-source compiler that finds task parallelism in C/C++ programs with no human intervention. Challenges included how to estimate the profit of tasks, bound their recursive depth and limit memory regions onto which they operate. 2.5-year project.
- Python | LLVM | C/C++ | Bash | Static Analyses | Parallel Computing | OpenMP

EDUCATION

MSc., Computer Science

Federal University of Minas Gerais (UFMG)

2016 - 2018 Brazil

BSc., Software Engineering (1 year)

University of East London (UEL)

2013 - 2014 United Kingdom

BSc., Computer Science

Federal University of Minas Gerais (UFMG)

2011 - 2016 Brazil

SKILLS

imperative
parallel
functional
low-level

cursor, claude (sonnet) & AI tools
archlinux user Compilers
Parallelism Operating Systems

Disclaimer from the candidate:I advocate for the transference of skills and I do believe those skills below reflect my history and experience to fuel the development of new tech.

go, c, C++, Python
POSIX, bash, js, ts
Assembly, CUDA, LLVM, SQL

docker git & git flow
clean code & clean arch redis aws
gcc & clang OpenMP
SCRUM agile

LANGUAGES

English (Cambridge CPE)
Portuguese
German
Italian

EXPERIENCE - EDUCATION

Compilers, Operating S. & Parallel Computing Professor - PUC Minas University

📅 Sep 2023 – Today

Summer Professor - Éstiam - École supérieure des technologies de l'information appliquées aux métiers - Paris

📅 Aug 2023

- One week hands-on course on Golang and Concurrency Patterns. ESTIAM University in Paris.

Software Specialist - Trybe

📅 Jul 2021 – May 2022

- Part-time as Software Specialist at Trybe, taught groups of up to 150 students remotely in live coding sessions.
- NodeJS, Express, Infra, Docker, HTTP, REST, JWT, ORM, SQL, Mongo, Deploy, Heroku, Github Actions, and others.

Algorithms Tutor

📅 Aug 2017 – Today

- Teaching algorithms privately for students since 2015. Data structures, algorithm complexity, databases, programming languages, compilers and algorithm analysis.

Research Assistant - Compilers Laboratory at UFMG (LAC)

📅 Sep 2015 – Feb 2018

- Carried out academic activities such as tutoring, teaching, researching, participating in seminars of 4 papers/week. Worked with a new pointer disambiguation analysis and acquired knowledge in compilers' data-flow static analyses and optimizations.

Algorithms and Data Structures II Professor - UFMG

📅 Aug 2017 – Dec 2017

- Sorting | Trees | Queues | Stacks | Graphs

Software Developer Intern - Synergia

📅 Feb 2015 – Jul 2015

- Java Spring | Hibernate | JUnit | Cucumber | Vaadin | Strong TDD Culture | Pair Programming

Software Developer Intern - JPF

📅 July 2014 – Jan 2015

- PHP | Yii Framework | MySQL

PUBLICATIONS

★ **Annotation of Tasks in Structured Code** RAMOS, P., SOUZA, G., DIVINO, C., ARAUJO, G., LEOBAS, G. and PEREIRA, F. PACT, 2018.

★ **TaskMiner: Identificação Automática de Tarefas.** RAMOS, P., SOUZA, G., LEOBAS, G. and PEREIRA, F. SBLP, 2018.

★ **Brazilian news: an analysis on the polarity of the headlines.** RAMOS, P., REIS, J. and BENEVENUTO, F. BraSNAM, 2016.

Pointer Disambiguation via Strict Inequalities. MAALEJ, M., GONNORD, L., PEREIRA, F., PAISANTE, V. and RAMOS, P. CGO, 2017.

Queer Identities in Video Games: Data Visualization for a Quantitative Analysis of Representation. BRAGANÇA, L., UTSCH, S., RAMOS, P., TENORIO, J. and CALDEIRA, P. SBGames, Brazil. 2017.

PROJECTS

Taskminer:

A tool for automatic mining of task parallelism in C programs. Written in C++, inside the LLVM infrastructure.

Visualizing the LGBTQ VG Archive:

Data visualization framework for visualizing the LGBTQ VG Archive. Available at <http://homepages.dcc.ufmg.br/~pedroramos/LGBTQ/page/>.

Quantifying Contamination in Programs:

Aliasing-sensitive algorithm to quantify contamination in programs by tainted flow analysis.

Tainted Flow Analysis in Smart Contracts:

Control flow graph static analysis that finds tainted paths from inputs to secrets on smart contracts. Secrets are methods that execute external functions without being sanitized, for example.

EAGER TO LEARN

Cairo/Solidity